Number 9 — Comparative costs · Taxation · Salaries and income tax · Foreign currency

Calculator allowed. Marks in brackets. ★ = harder question. Money answers to the nearest cent/penny unless told otherwise.

1 Which car has the better fuel efficiency? Show your working. [4]

car	fuel used	distance
Vega	9.5 litres	150 km
Orion	13.2 litres	220 km

> Compare kilometres per litre: Vega $150 \div 9.5 = 15.79$ km/l, Orion $220 \div 13.2 = 16.67$ km/l. Orion travels further on each litre, so Orion is more efficient. (Litres per 100 km gives the same conclusion: 6.33 vs 6.00.)

2 A phone costs £552 including sales tax at 20%. Calculate the price before tax. [3]

> The price with tax is 120% of the original, so original $= 552 \div 1.2 = £460.00$. (Not $552 - 20\%$ of 552, which would give £441.60.)

3 The country of Numeria has these income-tax rates. [8]

tax rate	annual salary
0%	first \$10 000
20%	\$10 001 to \$40 000
40%	above \$40 000

Mr Costa earns \$36 000 a year and Mrs Costa earns \$58 000 a year. How much tax does the family pay per week?

> Each band is taxed only on the money *inside* it.
>
> Mr Costa: 0 on the first $10\,000$; 20% of the remaining $26\,000 = \$5\,200$.
>
> Mrs Costa: 0 on $10\,000$; 20% of the next $30\,000 = \$6000$; 40% of the $18\,000$ above $40\,000 = \$7200$. Total $\$13\,200$.
>
> Family per year $5200 + 13\,200 = \$18\,400$; per week $18\,400 \div 52 = \$353.85$.

4 £1 (UK pounds) $= \$1.28$ (US dollars) $= €1.17$ (euros). Convert [6]

a £40 into dollars **b** \$40 into pounds **c** €40 into dollars.

> **a** $40 \times 1.28 = \$51.20$. **b** $40 \div 1.28 = £31.25$. **c** Go through pounds: $40 \div 1.17 = £34.19$, then $\times 1.28 = \$43.76$.

5 Nadia wants to buy a flat in Spain costing €240 000 using UK pounds. In March the exchange rate was £1 = €1.15; in June it was £1 = €1.20. How much would she save, in pounds, by buying in June rather than March? [4]

> March: $240\,000 \div 1.15 = £208\,695.65$. June: $240\,000 \div 1.20 = £200\,000$. Saving $= £8\,695.65$ — a stronger pound buys more euros, so the same euro price costs fewer pounds.

Total: 25 marks

Check: best value = compare per unit (per litre, per 100 g, km per litre) · a price including tax is (100 + rate)% of the original — divide, don't subtract · tax bands: only the slice inside each band is taxed at that rate · currency: multiply to go from the £1 side, divide to come back · convert via the common currency when neither is £.